

Sustainability Assessment Report

Overall Strategy Confidence Score: 96.0/100.0

1. Executive Summary

This assessment outlines the operational carbon footprint and provides a strategic, financially structured sustainability roadmap. Transitioning to rooftop solar PV and smart LED retrofitting are identified as high-yield initiatives.

2. Carbon Footprint Analysis

Organizational annual greenhouse gas footprint is calculated at 35.2 metric tons of CO2e. Grid electricity consumption represents 80.5% of total emissions.

3. Recommended Sustainability Measures

Measure Title	SDG Alignment
Solar Photovoltaic Installation (Expected 30% reduction, SDG 7)	SDG 12, SDG 13
LED Lighting Retrofit (Expected 10% reduction, SDG 12)	SDG 12, SDG 13
Water Recycling System (Expected 5% reduction, SDG 12)	SDG 12, SDG 13

4. Financial Analysis & Payback Schedule

Total required capital spending is \$25,000, yielding annual energy bill savings of \$6,500 with an average ROI of 26% and payback period of 3.8 years.

5. Phased Implementation Roadmap

Phased 12-month implementation timeline starts with quick-win LED Retrofits (0-3 Months), Greywater filtration systems (3-6 Months), and rooftop Solar PV installation (6-12 Months).

6. Immediate Strategic Next Steps

- Obtain local utility grid approvals for Solar PV.
- Gather supplier quotes for building-wide LED lighting retrofit.
- Establish employee awareness workshop on waste reduction.